



Selecting TDS Calibration Solutions

A Total Dissolved Solids (TDS) standard calibration solution should contain the same types of dissolved solids known to be water sample. Failure to do so may result in significant discrepancies between the ppm reading and the actual ppm of TDS in the test solution.

The easiest way to obtain a TDS standard calibration solution is to use a ready-made standard solution that has the same types of dissolved solids as the solution to be tested.

They are commonly formulated with either potassium chloride (KCl) or a 442 natural water formulation (40% sodium sulphate, 40% sodium bicarbonate and 20% sodium chloride).

Choose a ready-made calibration solution according to the following guidelines:

- Choose a TDS standard calibration solution that contains the same types of dissolved solids to be tested.
- Choose a TDS standard calibration solution that has a ppm value as close as possible to the value of the solution to be tested. If this is not reasonable because of the variations in the test solution, it is best to calibrate the TDS standard that has a ppm value in the upper one third of the TDS indicator's measurement range.
- Overleaf is a list of ppm TDS standard calibration solutions available for Palintest Instruments, with their contents and applications described. Standardization values of the conductivity calibration solutions are based on conditions of 25°C.

Note TDS meters are like all other dissolved solids testers in that they use the detected conductivity of a solution to give a direct reading in ppm (parts per million) or ppt (parts per thousand) of TDS.

The relationship between Conductivity and TDS is greatly affected by the types of dissolved solids or salts present in the solution. It is absolutely necessary to use a calibration solution similar in dissolved solids content to the solution to be tested in order to get acceptable accuracy. Also it is recommended that the calibration solution be at the same temperature as the test solution to minimize.

Application	Solution Part Code	Use TDS value
TDS of potassium chloride solutions & general purpose TDS measurements	PT 142/2 – 12.88mS/cm KCl solution	7447ppm (F = 0.578)
	PT 142/3 – 1413 μ S/cm KCl solution	744ppm (F = 0.527)
	PT 142/4 - 74 μ S/cm KCl solution	37ppm (F = 0.504)
TDS of saline swimming pools	PT 142/8 – 3000ppm 442 solution	2010ppm (F = 0.500)
TDS of lakes & streams, boiler water & cooling towers, general water treatment and swimming pools	PT 142/8 – 3000ppm 442 solution	3000ppm
	PT 142/9 – 300ppm 442 solution	300ppm
	PT 142/10 – 50ppm 442 solution	50ppm

* = please note, all values above are defined at 25°C.